Adnan Shahriar, PhD

Phone: 210-803-5230 | Email: shahriar.adnan@yahoo.com
LinkedIn: https://www.linkedin.com/in/adnan-shahriar/
GitHub: https://github.com/AdnanShahriar64
Google Scholar: https://scholar.google.com/citations?user=jF2HLEcAAAAJ

PROFESSIONAL SUMMARY

Mechanical engineering Ph.D. with experience in CFD-driven HVAC product development, numerical solver development, aerodynamic analysis, and experimental validation. Developed parametric design automation and improved outdoor fan-shroud airflow efficiency through simulation and prototype testing.

TECHNICAL SKILLS

Simulation: CFD, Finite Element Method, Fluid-Structure Interaction, Thermal-Fluid Modeling
Programming and CAD: Python, C++, MATLAB, CadQuery, SOLIDWORKS

PROFESSIONAL EXPERIENCE

Project Engineer — Rheem Manufacturing (Friedrich Air Conditioning) | San Antonio, TX *Mar 2025–Present*

- Optimized outdoor fan-shroud geometry using CFD; prototype tests demonstrated more than 80% improvement in airflow efficiency (CFM/W) relative to the previous design.
- Developed a Python/CadQuery parametric fan-shroud generator, automating CAD geometry and STEP-file creation across product configurations.

Senior Simulation Integration Engineer — L&T Technology Services | Plano, TX *Jan 2024–Mar 2025*

- Developed a real-time C/C++ HVAC plant model for SiL and HiL applications, coupling FEM-based fluid and thermal solvers with heater and cooler models.

Graduate Researcher — Resilient Extraterrestrial Habitats Institute | San Antonio, TX *Dec 2019–Jan 2024*

- Developed Eulerian-based CFD models of hurricane-wave generation and fluid motion to investigate hydrodynamic loading on coastal bridges.
- Developed coupled Eulerian-Lagrangian models to analyze fluid-structure interaction under extreme coastal loading.

Other relevant experience

EDUCATION

Ph.D., Mechanical Engineering — University of Texas at San Antonio, 2023 | GPA: 3.94
B.S., Aeronautical Engineering (Aerospace Major) — Military Institute of Science and Technology, 2015

SELECTED TECHNICAL DEMONSTRATIONS

CFD and Bridge Structural Analysis: https://youtu.be/DTQD6CamibM
HVAC Duct CFD: https://www.linkedin.com/feed/update/urn:li:activity:7258219618144538624/